

PlaceMux Phase 3 - Task 3

Performance Profiling & Bottleneck Elimination

Data Analyst | Sprint A - Scale & Reliability

Validation	PASS
Real source	UCI Online Retail transaction workbook
Raw rows	541,909
Clean rows	397,884
Gross revenue	GBP 8,911,407.90
Best measured p95 improvement	99.9%
Revenue sensitivity range	GBP 8,911.48 - GBP 44,557.41

Scope note: the supplied brief requires real production data, but no PlaceMux production performance/session telemetry was supplied. This report therefore uses the real UCI Online Retail transaction source as an explicit external proxy. Performance-to-conversion figures are not presented as observed PlaceMux production behavior. The financial section is a sensitivity model.

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1. Task objective and analytical bar

The task focuses on user-perceived system performance: whether speed and errors cost money. Its required outputs are an analysis linking latency/errors to conversion and abandonment, quantified revenue impact, and a prioritized list showing where speed is worth engineering effort. The supplied brief also requires real-data analysis, validation, a visible correlation-with-limits view, and a deliberate failure path. ■filecite■turn28file0■

The implementation meets the performance and commercial framing while deliberately blocking a causal claim. Because the available UCI transaction source does not contain session-level latency, request errors, conversion or abandonment events, an observed latency-to-conversion relationship cannot be calculated honestly from this source alone.

2. Data source and methodology

The analysis uses **data/online_retail/Online Retail.xlsx**. The source contains 541,909 raw transaction lines; the analysis uses 397,884 clean positive-purchase rows and gross revenue of GBP 8,911,407.90. The benchmark workloads are customer order retrieval, multi-customer summary, and revenue aggregation.

For each workload, baseline and optimized p95 latency are compared. The engineering changes represented by the benchmark are indexed access, query-shape optimization (including the N+1-to-grouped-query pattern), and reduced database work. The implementation also checks authorization, cross-tenant isolation, idempotent retry and persistence.

3. Performance benchmark results

Workload	Before p95	After p95	Improvement
GET /api/customer/orders	19.45 ms	0.02 ms	99.9%
GET /api/customers/summary	201.23 ms	0.19 ms	99.9%
GET /api/revenue	434.23 ms	71.33 ms	83.6%

The strongest measured result is the customer order and customer-summary workloads, each showing a 99.9% p95 reduction. Revenue aggregation improved by 83.6% at p95. These are benchmark comparisons from the local implementation, not PlaceMux production SLOs.

Engineering interpretation

- **Customer order retrieval:** indexed customer access removes avoidable scanning work.
- **Customer summary:** grouped/combined query logic eliminates repeated per-customer database work.
- **Revenue aggregation:** date-aware access reduces the cost of repeated aggregation across the transaction table.

4. Commercial impact sensitivity

The real transaction source provides 18,532 clean invoices and a gross-revenue base of GBP 8.91M. The sensitivity model asks: if a performance problem were associated with a 0.1, 0.25 or 0.5 percentage-point conversion loss, what revenue exposure would that imply at the observed average order value?

Conversion sensitivity	Implied transactions	Revenue exposure
0.10 pp	18.53	GBP 8,911.48
0.25 pp	46.33	GBP 22,278.71
0.50 pp	92.66	GBP 44,557.41

Critical interpretation: these are scenario exposures, not observed lost revenue. The source does not identify which users experienced latency or errors and cannot establish that performance caused conversion loss. The supplied task explicitly warns against claiming causation from observational correlation. █filecite█turn29file0█

5. Engineering prioritization

Priority	Fix	Evidence	Financial rationale
P0	Customer-order access/indexing	99.9% p95 improvement	Protects every request in a high-frequency access path; test against production conversion telemetry before monetizing.
P0	Customer-summary query rewrite	99.9% p95 improvement	Eliminates repeated DB work and has the same high sensitivity to user-facing performance.
P1	Revenue aggregation optimization	83.6% p95 improvement	Large tail-latency reduction; prioritize after the highest-impact user-facing paths.

6. Validation, security and failure controls

Control	Result
Happy path	PASS
Unauthorized request	PASS
Cross-tenant attack	PASS
Idempotent retry	PASS
Persistence check	PASS
Benchmark on real source	PASS
Causal claim blocked	PASS

The implementation intentionally includes a failure-path/guardrail posture: access controls are tested, cross-tenant requests are rejected, retries are checked for idempotency, and the analytical model blocks a causal performance-to-conversion claim when the source cannot support one.

7. Traceability, limitations and production hand-off

- **Source:** data/online_retail/Online Retail.xlsx
- **Implementation:** phase3/task3_performance_money.py
- **Outputs:** task3_summary.json, before_after_p95.csv, financial_sensitivity.csv, task3_performance_money_dashboard.html
- **Primary limitation:** no session/request latency, error, conversion or abandonment telemetry exists in the real transaction source.
- **Production next step:** join request latency/errors, session abandonment, conversion and order value through a common session/request identifier, then re-estimate the performance-response curve with confidence intervals or an experiment.
- **Hand-off:** Frontend/Backend should prioritize the two 99.9%-improvement access paths first, while Analytics/Instrumentation should add the missing telemetry needed to attach observed rupees to observed performance cohorts.

Final conclusion

The benchmark demonstrates substantial latency reduction from concrete engineering changes. The commercial sensitivity model shows that even small conversion shifts can correspond to thousands of pounds in revenue exposure on this real transaction base. However, the evidence does not justify claiming that the measured latency caused conversion loss. The next production-grade step is instrumentation that links performance cohorts to user outcomes.